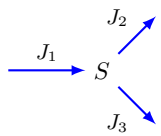
1 Simple Branch Structure

1.1 Basic Topology



1.2 Mass Conservation

At any branch node with b branches entering and d leaving:

$$\sum_{i=1}^{b} v_i - \sum_{j=1}^{d} v_j = \frac{dS}{dt}$$

Steady-state constraint:

$$J_1 = J_2 + J_3$$

1.3 Flux Fraction

Define the flux split:

$$\alpha = \frac{J_2}{J_1} \quad \text{(fraction through } J_2\text{)}$$

$$1 - \alpha = \frac{J_3}{J_1} \quad \text{(fraction through } J_3\text{)}$$

2 Control Coefficients

2.1 Counting Coefficients

For a simple branch with 3 enzymes and 1 intermediate:

Type	Count
Flux CCs	$3 \times 3 = 9$
Conc CCs	$3 \times 1 = 3$
Total	**12**

2.2 Control Coefficient Matrix

	e_1	e_2	e_3
J_1	$C^{J_1}_{e_1}$	$C^{J_1}_{e_2}$	$C^{J_1}_{e_3}$
J_2	$C^{J_2}_{e_1}$	$C^{J_2}_{e_2}$	$C^{J_2}_{e_3}$
J_3	$C^{J_3}_{e_1}$	$C^{J_3}_{e_2}$	$C^{J_3}_{e_3}$
S	$C^{S}_{e_1}$	$C^{S}_{e_2}$	$C^{S}_{e_3}$

3 Summation Theorems

Each flux has its own summation theorem:

$$C^{J_1}_{e_1} + C^{J_1}_{e_2} + C^{J_1}_{e_3} = 1$$
$$C^{J_2}_{e_1} + C^{J_2}_{e_2} + C^{J_2}_{e_3} = 1$$
$$C^{J_3}_{e_1} + C^{J_3}_{e_2} + C^{J_3}_{e_3} = 1$$

And for concentration:

$$C^{S}_{e_1} + C^{S}_{e_2} + C^{S}_{e_3} = 0$$

4 Connectivity Theorems

One connectivity theorem per flux:

$$C^{J_1}_{e_1}\varepsilon^{v_1}_S + C^{J_1}_{e_2}\varepsilon^{v_2}_S + C^{J_1}_{e_3}\varepsilon^{v_3}_S = 0$$
$$C^{J_2}_{e_1}\varepsilon^{v_1}_S + C^{J_2}_{e_2}\varepsilon^{v_2}_S + C^{J_2}_{e_3}\varepsilon^{v_3}_S = 0$$
$$C^{J_3}_{e_1}\varepsilon^{v_1}_S + C^{J_3}_{e_2}\varepsilon^{v_2}_S + C^{J_3}_{e_3}\varepsilon^{v_3}_S = 0$$

Problem: 3 unknowns, only 2 equations (summation + connectivity). Need a third equation!

5 Branch Point Theorems

5.1 Derivation Strategy

Thought experiment:

1. Perturb e_2 by δe_2
2. Compensate with e_3 such that $\delta S = 0$
3. Since $\delta S = 0$ and e_1 unchanged: $\delta J_1 = 0$
4. Use $\delta v_2 + \delta v_3 = 0$ at branch point

5.2 Flux Branch Point Theorems

With respect to J_1:

$$C^{J_1}_{e_2}(1-\alpha) - C^{J_1}_{e_3}\alpha = 0$$

With respect to J_2:

$$C^{J_2}_{e_1}(1-\alpha) + C^{J_2}_{e_3} = 0$$

With respect to J_3:

$$C^{J_3}_{e_1}\alpha + C^{J_3}_{e_2} = 0$$

5.3 Concentration Branch Point Theorems

$$C^{S}_{e_2}(1-\alpha) - C^{S}_{e_3}\alpha = 0$$

Variants:

$$C^{S}_{e_1}(1-\alpha) + C^{S}_{e_3} = 0$$
$$C^{S}_{e_1}\alpha + C^{S}_{e_2} = 0$$

6 Solving for Control Coefficients

6.1 Matrix Formulation

Combine summation, connectivity, and branch theorems:

$$\mathbf{C} \cdot \mathbf{T} = \mathbf{I}$$

For J_2:

$$\begin{bmatrix} C^{J_2}_{e_1} & C^{J_2}_{e_2} & C^{J_2}_{e_3} \end{bmatrix} \begin{bmatrix} 1 & \varepsilon_1 & 0 \\ 1 & \varepsilon_2 & 1-\alpha \\ 1 & \varepsilon_3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

6.2 Explicit Formulas for J_2

Let $D = \varepsilon_2\alpha + \varepsilon_3(1-\alpha) - \varepsilon_1$ (denominator).

Note: $D > 0$ when $\varepsilon_1 < 0$, $\varepsilon_2, \varepsilon_3 > 0$.

Flux control coefficients:

$$C^{J_2}_{e_1} = \frac{\varepsilon_2}{D} > 0$$

$$C^{J_2}_{e_2} = \frac{\varepsilon_3(1-\alpha) - \varepsilon_1}{D} > 0$$

$$C^{J_2}_{e_3} = \frac{-\varepsilon_2(1-\alpha)}{D} < 0$$

Concentration control coefficients:

$$C^{S}_{e_1} = \frac{1}{D} > 0$$

$$C^{S}_{e_2} = \frac{-\alpha}{D} < 0$$

$$C^{S}_{e_3} = \frac{-(1-\alpha)}{D} < 0$$

6.3 Sign Patterns

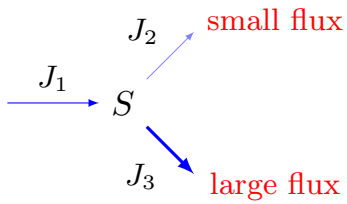
	e_1	e_2	e_3
C^{J_2}	+	+	−
C^{S}	+	−	−

Key insight: $C^{J_2}_{e_3} < 0$ means **competition** between branches—increasing e_3 "steals" flux from J_2.

7 Branch Point Ultrasensitivity

7.1 Case 1: Most Flux Through J_3

When $\alpha \to 0$ (small flux through J_2):



As $\alpha \to 0$:

$$C^{J_2}_{e_2} \to 1$$

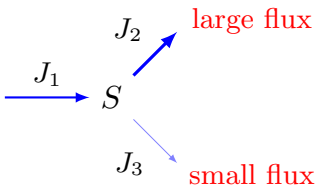
$$C^{J_2}_{e_3} \to \frac{\varepsilon_2}{\varepsilon_1 - \varepsilon_3}$$

Ultrasensitivity: Control coefficients can **greatly exceed 1**!
Example: $C^{J_2}_{e_1} = 8.34$ means 1% increase in e_1 causes 8% increase in J_2.

Analogy: A small stream off a large river—flooding in the river has huge impact on the stream.

7.2 Case 2: Most Flux Through J_2

When $\alpha \to 1$ (large flux through J_2):



As $\alpha \to 1$:

$$C^{J_2}_{e_2} \to \frac{\varepsilon_1}{\varepsilon_1 - \varepsilon_2}$$

$$C^{J_2}_{e_3} \to 0$$

Pathway behaves as a **linear chain** ($J_1 \to J_2$). The minor branch J_3 has negligible influence.

8 Conditions for Ultrasensitivity

8.1 Requirements

To maximize sensitivity of J_2 at a branch:

1. **Asymmetric flux:** Most flux through J_3 ($\alpha \ll 1$)
2. **High ε_2 relative to ε_3:**
   - Positive cooperativity at E_2
   - v_3 more saturated than v_2 ($K_m^{(2)} > K_m^{(3)}$)
3. **Small product inhibition:** $|\varepsilon_1|$ small

8.2 Extreme Case

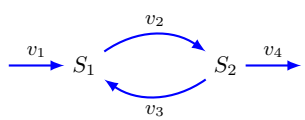
It is possible to arrange kinetics such that:

- $C^{J_2}_{e_1} = +1$
- $C^{J_2}_{e_2} = +1$
- $C^{J_2}_{e_3} = -1$

Every step is equally "rate-limiting"! This shows rate limitation is not a simple concept.

9 Futile (Substrate) Cycles

9.1 Structure



Flux constraint:

$$v_1 = v_2 - v_3 = v_4$$

Examples: Glucose ↔ G6P, F6P ↔ F1,6BP

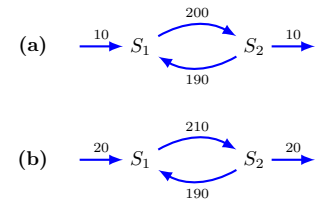
9.2 Amplification Mechanism

Maximum amplification:

$$\frac{\delta v_1/v_1}{\delta v_2/v_2} = \frac{v_2}{v_1} = 1 + \frac{v_3}{v_1}$$

Higher cycling rate (v_3) relative to through flux (v_1) gives greater amplification.

9.3 Numerical Example



5% increase in $v_2 \Rightarrow$ **100% increase** in v_1!
(Assuming v_3 unchanged due to saturation)

9.4 MCA Analysis of Cycles

Flux control coefficient for J_1 with respect to e_2:

$$C^{J_1}_{e_2} = \frac{\varepsilon_1^1 \varepsilon_2^4 (1 + v_3/v_1)}{D}$$

Key requirements:

1. Product inhibition on v_1 ($\varepsilon_1^1 \neq 0$) is **essential**
2. Small ε_2^3 (weak activation of reverse arm)

9.5 Condition for High Sensitivity

When cycling rate $\gg$ through flux:

$$C^{J_1}_{e_2} \approx \frac{v_2}{v_3 \frac{\varepsilon_2^3}{\varepsilon_2^4} - v_2 \frac{\varepsilon_1^2}{\varepsilon_1^1}}$$

Maximal when:

$$\frac{\varepsilon_2^3}{\varepsilon_2^4} + \frac{\varepsilon_1^2}{\varepsilon_1^1} \ll 1$$

Meaning:

- S_2 activates v_4 more than v_3
- S_1 inhibits v_1 more than it activates v_2

10 Summary

Feature	Key Point
Branch theorem	Third equation needed to solve system
Competition	$C_{e_j}^{J_i} < 0$ for competing branch
Ultrasensitivity	$\lvert C\rvert \gg 1$ when minor branch observed
Linear limit	$\alpha \to 1$: pathway becomes chain
Futile cycles	Amplification $= 1 + v_3/v_1$

Based on: H.M. Sauro, "Systems Biology: An Introduction to Metabolic Control Analysis", March 19, 2026